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https://www.tensorflow.org/api_docs/python/tf/VariableAggregation

Indicates how a distributed variable will be aggregated.

Function Signatures

```
var.assign_add(x)
```

Code Examples

```
strategy = tf.distribute.MirroredStrategy(["GPU:0", "GPU:1"])
with strategy.scope():
    v = tf.Variable(5.0, aggregation=tf.VariableAggregation.MEAN)
    @tf.function
    def update_fn():
        return v.assign_add(1.0)
    strategy.run(update_fn)
    PerReplica:{
    0: ,
    1:
    }
```

```
strategy = tf.distribute.MirroredStrategy(["GPU:0", "GPU:1"])
```

```
with strategy.scope():
```

```
v = tf.Variable(5.0, aggregation=tf.VariableAggregation.MEAN)
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@tf.function
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def update_fn():
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    return v.assign_add(1.0)
```

```
    strategy.run(update_fn)
```

Documentation

Indicates how a distributed variable will be aggregated.

`tf.distribute.Strategy` distributes a model by making multiple copies (called "replicas") acting on different elements of the input batch in a data parallel model. When performing some variable-update operation, for example `var.assign_add(x)`, in a model, we need to resolve how to combine the different values for `x` computed in the different replicas.

- `NONE`: This is the default, giving an error if you use a

variable-update operation with multiple replicas.

- `SUM`: Add the updates across replicas.
- `MEAN`: Take the arithmetic mean ("average") of the updates across replicas.
- `ONLY_FIRST_REPLICA`: This is for when every replica is performing the same

update, but we only want to perform the update once. Used, e.g., for the global step counter.

For example:

`## Class Variables`